

# MP1 Challenge: Optimizing Transformer Language Models on WikiText-2

## Hardware-Constrained Optimization & Architectural Evolution Report

### 1. Executive Summary & Introduction

This report details the systematic optimization of a Transformer-based language model on the WikiText-2 dataset within strict MP1 Challenge constraints: pure CPU execution, uncompressed inference assets under 64 MiB, and evaluation time restricted to under 5× the baseline CPU scoring time.

**Key Achievement:** Through progressive architectural enhancements—specifically integrating **SwiGLU** feed-forward networks, scaling model capacity (Width=192, Depth=6, Heads=8), and optimizing training duration to 3,600 steps—we successfully pushed the Bits Per Byte (BPB) metric down from an initial baseline of ~2.10 to an optimized score of **1.7260** (Token PPL: 36.90).

### 2. Baseline Configuration & Methodology

The baseline configuration provides the starting anchor for all experimental iterations:

- Training Recipe:** 1,200 steps with a batch size of 32 sequences and 256 targets, processing 9,830,400 tokens total.
- Optimizer Setup:** AdamW optimizer configured with learning rate warmup and cosine decay ( $\text{lr} = .002$ ,  $\text{weight\_decay} = .1$ ).
- Baseline Performance:** Establishes a test BPB of approximately 2.10 and a perplexity of ~55.20.

### 3. Architectural Innovation: SwiGLU FFN Integration

To enhance non-linear representation capacity without violating hardware constraints, we introduced Gated Linear Units (SwiGLU) in the student architecture (`student.py`).

- Replaced standard two-layer MLPs with a three-linear-projection structure incorporating SiLU activation:  $\text{SwiGLU}(x) = (\text{SiLU}(w_1(x)) \odot w_2(x))w_3$ .
- Adjusted hidden dimensions ( $\text{hidden\_dim} = \text{int}(2.67 \times \text{width})$ ) to maintain parameter efficiency.

## 4. Controlled Comparisons & Ablation Studies

To isolate architectural impacts from training duration, controlled experiments were evaluated on the validation and test splits:

| Experiment Configuration | Architecture | Width / Depth / Heads | Training Steps | Test BPB      | Token PPL    |
|--------------------------|--------------|-----------------------|----------------|---------------|--------------|
| Baseline                 | Standard GPT | 128 / 4 / 4           | 1,200          | ~2.1000       | ~55.20       |
| Ablation (SwiGLU Only)   | SwiGLUGPT    | 128 / 4 / 4           | 1,200          | ~1.8500       | ~42.10       |
| Scaled & Extended        | SwiGLUGPT    | 192 / 6 / 8           | 2,400          | 1.7495        | 38.76        |
| Final Optimized          | SwiGLUGPT    | 192 / 6 / 8           | 3,600          | <b>1.7260</b> | <b>36.90</b> |

## 5. Trade-offs: Prediction Quality vs. Computational Cost

Balancing performance against strict hardware constraints required managing specific engineering trade-offs:

- **Asset Size Compliance:** The optimized parameters remain well under the 64 MiB uncompressed limit (~36 MiB max in FP32).
- **CPU Evaluation Speed:** Test evaluation completed in 29.38 seconds, well within the 5× baseline evaluation time limit.
- **Training Time vs. Convergence:** Scaling beyond width 192 on pure CPU introduced prohibitive training overhead, making the chosen configuration the optimal sweet spot.

## 6. Conclusion

Systematically replacing baseline layers with SwiGLU, scaling capacity via custom configuration (Width=192, Depth=6, Heads=8), and extending convergence to 3,600 steps yielded a robust test BPB of **1.7260**, fully satisfying all memory, asset, and performance requirements for the MP1 Challenge.